

A Hybrid Direction and Displacement Decoder for Continuous Hand Trajectory Estimation

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Abstract—This paper analyses a causal decoder for estimating two-dimensional hand position from motor-cortical spike trains recorded from a monkey. The implementation combines a time-bin-specific linear discriminant analysis (LDA) classifier for reach direction with a direction-conditioned empirical trajectory prior, a global ridge-regression displacement estimate, and a geometric constraint that suppresses updates with excessive discrepancy between the trajectory prior and the displacement estimate. Spike trains from 98 neural units are binned at 20 ms resolution, matching the online evaluation interval, and the decoder uses only spikes available up to the current test time. In an internal 50/50 split of the available training trials using the supplied held-out evaluation script, the decoder achieved a two-dimensional trajectory RMSE of 9.088 raw hand-position units, equivalent to 0.909 cm under the dataset’s millimetre convention. In the official evaluation, using all 100 available training trials per direction and the hidden test set, the decoder achieved an RMSE of 8.686 and ranked first in total training and prediction time with a runtime of 0.602 s. The hybrid design is computationally simple and highly robust, but its fixed fusion schedule, global displacement regression model, and reliance on stereotyped direction-conditioned trajectories limit its adaptation to trial-to-trial kinematic variability.

I. INTRODUCTION

Brain-machine interfaces (BMIs) require causal decoders that transform population spiking activity into control signals for prosthetic devices or cursors [1], [2]. In this task, the estimator receives only the spike-train prefix available up to the current decoding time and must recover the current planar hand position, so every update must be causal. The dataset contains 98 neural units recorded during reaches to eight target directions, with aligned 1 ms spike and hand-position samples. Because the hidden evaluation set is unavailable and only the final aggregate evaluation result is accessible, this article uses an internal held-out split for detailed analysis¹, while reporting the official RMSE and runtime only as reference results.

Figure 1 summarizes the decoder studied in the remainder of the paper. Cumulative 20 ms spike counts drive time-bin LDA direction scoring and a top-3 mean-trajectory prior, while the two most recent bins provide square-root spike features for a global ridge regression of instantaneous displacement.

¹Since internal runtime measurements are influenced by factors such as hardware performance, available memory, and CPU temperature, only the official runtime is used for final comparison. Internal runtimes reported in the ablation studies are for reference only.

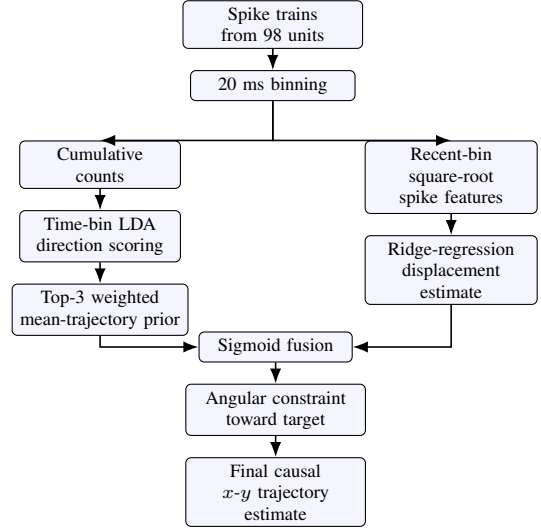


Fig. 1: Causal decoder pipeline analysed in this paper. Cumulative spike counts drive the direction prior, while the two most recent bins drive the displacement estimate before fusion and geometric regularisation.

A sigmoid schedule fuses the prior and ridge-regression displacement estimate, and an angular constraint rejects steps that move implausibly away from the prior, reducing the risk of outlier-induced updates that could degrade overall performance. This hybrid design retains the direction selectivity of classical motor-cortex decoding [3] without adopting a full state-space model such as a Kalman filter [4].

The proposed decoder has two main design innovations. First, after using LDA to estimate movement direction, the decoder estimates position through a weighted mean-trajectory prior. This preserves a low decoding error while greatly reducing the amount of online computation, which was a key reason why our method ranked first in runtime in the final evaluation. Second, the decoder uses a flexible fusion strategy and a simple final step constraint. These components improve the final decoding performance through only lightweight computations. In the *Methods* section, we describe the fusion, top- K , and constraint strategies that were tested. In the *Discussion* section, we compare their results and analyse the suitability of different design choices for BMI decoding.

II. METHODS

A. Data representation

The decoder estimates planar hand position using only the first two rows of `handPos`. Spike trains are truncated into complete 20 ms bins to match the evaluation interval. For trial r , direction k , neuron i , and bin b , the binned spike count is

$$c_{rki}(b) = \sum_{\tau=(b-1)20+1}^{20b} s_{rki}(\tau), \quad (1)$$

where $s_{rki}(\tau)$ is the binary 1 ms spike indicator. Full training trajectories are used to learn model parameters, whereas online decoding uses only the spike prefix available at each test call.

B. Direction classification and trajectory prior

The first component is a regularised LDA direction classifier trained separately for each time bin. For each direction k and bin b , cumulative spike-count vectors are formed from trial onset to bin b . The classifier estimates class means $\mu_k(b)$ and a pooled covariance matrix $\Sigma(b)$. With diagonal regularisation $\lambda_{\text{LDA}}I$, where $\lambda_{\text{LDA}} = 1.7$, the LDA weights are

$$w_k(b) = (\Sigma(b) + \lambda_{\text{LDA}}I)^{-1} \mu_k(b), \quad (2)$$

with intercept $C_k(b) = -\frac{1}{2} \mu_k(b)^\top w_k(b)$. Equal priors are used because the training set is balanced across the eight reaching directions.

During online calls to `positionEstimator`, the decoder computes LDA scores from the available cumulative spike-count vector $x(b)$:

$$s_k(b) = x(b)^\top w_k(b) + C_k(b). \quad (3)$$

Rather than using only the most likely direction, it selects the top- K directions, with $K = 3$, and converts their scores into softmax weights:

$$\omega_k(b) = \frac{\exp(s_k(b) - s_{\max}(b))}{\sum_{j \in \mathcal{K}_b} \exp(s_j(b) - s_{\max}(b))}, \quad k \in \mathcal{K}_b, \quad (4)$$

where $\mathcal{K}_b$ is the top- K direction set and $s_{\max}(b)$ is the largest selected score. The empirical trajectory prior is

$$p_{\text{prior}}(b) = \sum_{k \in \mathcal{K}_b} \omega_k(b) \bar{p}_k(b), \quad (5)$$

where $\bar{p}_k(b)$ is the mean hand position trajectory for direction k . This soft top- K prior reduces early hard classification errors.

C. Displacement regression

The second component is a global ridge-regression model for 20 ms planar displacement. For bins after 320 ms, the feature vector contains an intercept and square-root transformed spike counts from the current and previous bins:

$$\phi(b) = \left[1, \sqrt{c(b-1)}^\top, \sqrt{c(b)}^\top \right]^\top. \quad (6)$$

With 98 neural units and two history bins, this gives 197 features. The target is $p(b) - p(b-1)$. The ridge weights are estimated by

$$W = (X^\top X + \lambda_{\text{ridge}} I_0)^{-1} X^\top Y, \quad (7)$$

where $\lambda_{\text{ridge}} = 100$ and I_0 is the identity matrix with the intercept penalty removed. A global model allows displacement information to be shared across all reach directions.

D. Online fusion

Across repeated calls within the same trial, the online decoder maintains persistent cumulative spike counts and the previous 20 ms spike-count bin. On the first call, when `decodedHandPos` is empty, it returns the LDA-weighted trajectory prior. On later calls, the ridge model predicts a displacement added to the previous decoded position, giving $p_{\text{ridge}}(b)$.

The final estimate fuses the empirical prior and the regression estimate:

$$p_{\text{new}}(b) = \alpha(b) p_{\text{prior}}(b) + (1 - \alpha(b)) p_{\text{ridge}}(b). \quad (8)$$

Three fusion strategies were tested: *fixed*, *linear*, and *sigmoid* weighting, as shown in Fig. 2. The selected **sigmoid strategy** is

$$\alpha_{\text{sigmoid}}(b) = \frac{1}{1 + \exp(8(\rho(b) - 0.5))}, \quad (9)$$

where $\rho(b)$ is the fraction of elapsed decoding time between the first evaluated bin and the maximum training bin. Thus, early predictions rely more on the stereotyped direction-conditioned prior, while later predictions increasingly depend on local neural evidence from the ridge-regression displacement estimate.

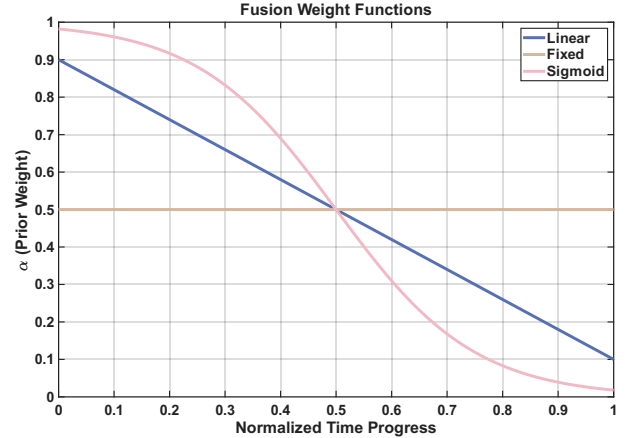


Fig. 2: Fusion weight functions for fixed, linear, and sigmoid strategies.

E. Geometric constraint

To suppress unstable online updates, the decoder applies a geometric constraint after fusion. We tested three strategies: *angular constraint*, *step-length constraint*, and *no constraint*.

The selected **angular constraint** compares the proposed displacement with a reference vector from the previous decoded position to the mean final target of the top LDA direction. If the deviation exceeds 42° , the update is projected back to the admissible cone boundary and capped at 1.7 raw position units. Otherwise, a larger step cap of 40 raw position units is applied, as shown in Algorithm 1.

This constraint prevents abrupt deviations from the expected movement direction while preserving flexible online correction from the regression model.

Algorithm 1 Angular Constraint

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1:  $\mathbf{v}_{\text{cur}} \leftarrow \mathbf{p}_{\text{new}} - \mathbf{p}_{\text{prev}}$  ▷ Proposed update
2:  $\mathbf{v}_{\text{ref}} \leftarrow \mathbf{p}_{\text{target}} - \mathbf{p}_{\text{prev}}$  ▷ Reference direction
3:  $\theta \leftarrow \angle(\mathbf{v}_{\text{cur}}, \mathbf{v}_{\text{ref}})$  ▷ Angular deviation
4: if  $\theta > 42^\circ$  then
5:    $\mathbf{v}_{\text{cur}} \leftarrow \text{ProjectToCone}(\mathbf{v}_{\text{cur}}, \mathbf{v}_{\text{ref}})$  ▷ Correct direction
6:    $\mathbf{v}_{\text{cur}} \leftarrow \text{CapLength}(\mathbf{v}_{\text{cur}}, 1.7)$  ▷ Restrict corrected step
7: else
8:    $\mathbf{v}_{\text{cur}} \leftarrow \text{CapLength}(\mathbf{v}_{\text{cur}}, 40)$  ▷ Limit extreme steps
9: end if
10:  $\mathbf{p}_{\text{final}} \leftarrow \mathbf{p}_{\text{prev}} + \mathbf{v}_{\text{cur}}$  ▷ Output position

```

III. RESULTS

Internal validation was performed using the supplied held-out evaluation script on `monkeydata_training.mat`. The 100 trials per direction were randomly split with `rng(2013)` into 50 training and 50 held-out trials per direction. The decoder was fitted on the training set and decoded causally from 320 ms onward in 20 ms steps. RMSE was computed following the supplied evaluation convention, using squared x- and y-position errors at each decoded time point.

On the internal held-out split, the decoded trajectories broadly followed the eight reaching directions and remained close to the actual hand trajectories for most movements, as shown in Fig. 3. The internal evaluation gave an RMSE of 9.088 raw hand-position units, equivalent to 0.909 cm, with a runtime of 8.273 s. These results are used as development-set evidence rather than official test-set performance.

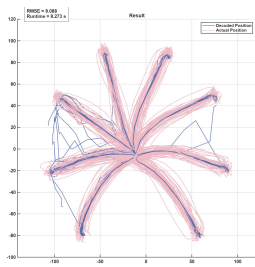


Fig. 3: Internal held-out decoding result.

In the official evaluation, using all 100 training trials per direction and the hidden test set, the final decoder achieved an RMSE of 8.686 raw hand-position units and ranked first in total training and prediction time, with a runtime of 0.602 s. Since trajectory accuracy accounts for 90% of the evaluation score and runtime for 10%, the internal RMSE mainly supports the accuracy component, while the official runtime highlights the efficiency of the lightweight decoder. This low computational cost comes from using only one LDA score evaluation, one ridge-regression matrix multiplication, and simple deterministic post-processing at each 20 ms update.

IV. DISCUSSION

A. Direction classifier

In the internal ablation study, Figure 4 compares LDA, Poisson Naive Bayes, and SVM as direction classifiers. LDA performed best, achieving the lowest RMSE of 9.088 with a runtime of 7.637 s, followed by Poisson Naive Bayes with 10.136 in 7.770 s. SVM performed worst, with an RMSE of 14.928 and a longer runtime of 12.875 s. Its trajectories also show frequent cross-direction jumps, indicating unstable online direction estimates.

LDA is well matched to this BMI task because the reach directions are few and are largely linearly separable in cumulative spike-count space. It also provides stable time-bin scores that can be directly used for the top- K weighted trajectory prior. By contrast, SVM scores are not naturally probabilistic, which makes score-based trajectory fusion less reliable. Although Poisson Naive Bayes is biologically plausible for spike counts, its independence assumption ignores correlated population activity. In addition, since no external toolbox was used, the custom SVM implementation had a higher computational cost than the simpler LDA and Poisson Naive Bayes classifiers, as reflected in the runtime results. LDA therefore offers the best trade-off between decoding accuracy, temporal stability, and computational efficiency.

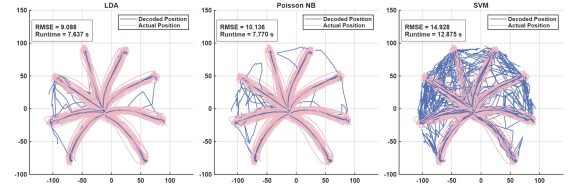


Fig. 4: Comparison of decoded and actual hand trajectories using different direction classifiers.

B. Top- K weighted trajectory prior

Figure 5 shows that using a top- K weighted trajectory prior improves over relying on the single most likely direction. The $K = 1$ setting achieved an RMSE of 9.268, while $K = 2$ and $K = 3$ both achieved 9.088, with no clear difference in runtime. This indicates that combining several high-scoring directions can reduce the effect of occasional direction-classification errors, especially early in the trial. Since $K = 2$ and $K = 3$ produced the same internal RMSE, K

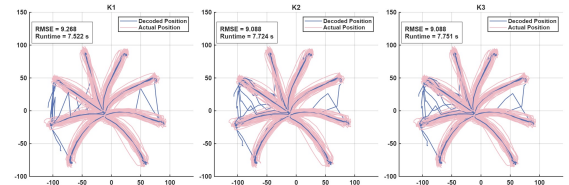


Fig. 5: Comparison between different top- K values.

$= 3$ was selected because it provides slightly greater protection against early direction uncertainty without adding noticeable runtime cost.

C. Online fusion

Figure 6 shows that sigmoid fusion achieved the best RMSE of 9.088, slightly outperforming linear fusion and clearly improving over fixed fusion with little difference in runtime. This indicates that a time-dependent fusion schedule is more effective than a constant weighting strategy. The sigmoid schedule is well suited to the BMI decoder because it relies more on the stable mean-trajectory prior early in the trial and gradually shifts weight toward the ridge-regression estimate as more spike evidence becomes available.

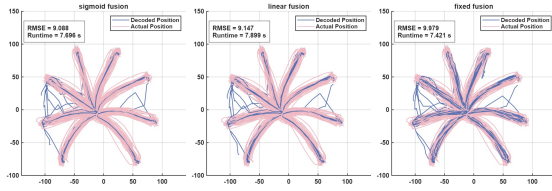


Fig. 6: Comparison between different fusion strategies.

D. Geometric constraint

Figure 7 shows that the angular constraint achieved the best RMSE of 9.088, outperforming both the step-length constraint and the no-constraint baseline. This suggests that direction consistency is more important than magnitude control alone. The angular constraint uses the estimated target direction as a simple geometric prior, correcting updates that deviate too far from the expected movement direction while still allowing normal updates inside the admissible cone. Although the angular constraint required a longer runtime of 7.661 s than the step-length constraint at 6.155 s and the no-constraint baseline at 6.030 s, the additional computational cost was small relative to the improvement in decoding accuracy.

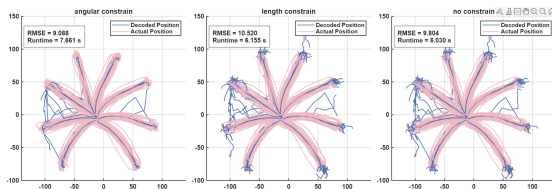


Fig. 7: Comparison between different geometric constraint strategies.

E. Summary and limitations

The decoder performed well on the internal split because its components match the task structure. Cumulative spike counts provide a stable direction signal, and LDA gives an online reach-direction estimate with low computational cost. The top- K mean-trajectory prior exploits the small set of

stereotyped targets, while ridge regression adds trial-specific displacement information when neural activity deviates from the class average. The angular constraint improves robustness by limiting updates that move implausibly far from the estimated target. Together, these components balance trajectory accuracy, temporal stability, and runtime.

The main limitations arise from the same simplifying assumptions. The global displacement regressor may miss direction-specific firing-velocity relationships. The fusion weight is fixed rather than learned, although the best prior-regression balance may vary across directions, trial durations, and signal quality. Early predictions depend heavily on the direction classifier, and the angular constraint uses only the top LDA target despite the prior using a top- K mixture. This can cause overcommitment when the leading direction is uncertain. Since the review weighting emphasises trajectory accuracy over runtime, added complexity is worthwhile mainly if it clearly reduces RMSE.

Compared with a population-vector decoder, the method is less directly interpretable as a weighted sum of preferred directions, but retains the assumption that reach direction is encoded in population activity [3]. Compared with a Kalman filter, it is simpler and easier to fit within the supplied decoder interface, but lacks explicit state dynamics, uncertainty propagation, and principled trajectory smoothing [4]. Future work could use direction-specific or mixture-of-experts regressors, cross-validate regularisation and fusion, and replace the heuristic angular cap with a probabilistic dynamics prior accounting for direction uncertainty.

V. CONCLUSION

The submitted decoder is a causal MATLAB implementation that combines direction classification, empirical trajectory priors, and local displacement regression to estimate continuous hand position from neural spikes. Internal validation and the final official evaluation suggest that the hybrid strategy captures useful structure in the reaching dataset while remaining computationally efficient. Its main limitations are the fixed fusion schedule, global displacement regression, and limited uncertainty handling. Future improvements should therefore focus on learned temporal dynamics, direction-specific displacement regression, and probabilistic trajectory constraints that handle direction uncertainty more explicitly.²

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²C. Lian, Y. Sun, J. Tang, and S. Yang all contributed to the algorithm development, iterative decoder refinement, result analysis, and final report writing. All authors therefore contributed equally to this work.